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Find the derivative of the function.

$$y = \frac{5x - 2}{4x + 3}$$

To find y' , use the Derivative Quotient Rule, which states if u and v are differentiable at x and if $v(x) \neq 0$, then the quotient $\frac{u}{v}$ is differentiable

$$\text{at } x, \text{ and } \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}.$$

$$y' = \frac{d}{dx} \left(\frac{5x - 2}{4x + 3} \right) = \frac{(4x + 3) \frac{d}{dx}(5x - 2) - (5x - 2) \frac{d}{dx}(4x + 3)}{(4x + 3)^2}$$

To find $\frac{d}{dx}(5x - 2)$, use the Derivative Sum Rule, which states if u and v are differentiable functions of x , then their sum $u + v$ is differentiable at

every point where u and v are both differentiable. At such points, $\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$.

$$\frac{d}{dx}(5x - 2) = \frac{d}{dx}(5x) - \frac{d}{dx}(2)$$

To find $\frac{d}{dx}(5x)$, use the Constant Multiple Rule and the Power Rule for Positive Integers.

The Constant Multiple Rule states if u is a differentiable function of x , and c is a constant, then $\frac{d}{dx}(cu) = c \frac{du}{dx}$.

The Power Rule for Positive Integers states if n is a positive integer, then $\frac{d}{dx}(x^n) = nx^{n-1}$.

$$\frac{d}{dx}(5x) = 5$$

To find $\frac{d}{dx}(2)$, use the Derivative of a Constant Function Rule, which states if f has the constant value $f(x) = c$, then $\frac{df}{dx} = \frac{d}{dx}(c) = 0$.

$$\frac{d}{dx}(2) = 0$$

$$\begin{aligned} \text{Thus, } \frac{d}{dx}(5x - 2) &= \frac{d}{dx}(5x) - \frac{d}{dx}(2) \\ &= 5 - 0 \\ &= 5. \end{aligned}$$

To find $\frac{d}{dx}(4x + 3)$, use the Derivative Sum Rule.

$$\frac{d}{dx}(4x + 3) = \frac{d}{dx}(4x) + \frac{d}{dx}(3)$$

To find $\frac{d}{dx}(4x)$, use the Constant Multiple Rule and the Power Rule for Positive Integers.

$$\frac{d}{dx}(4x) = 4$$

To find $\frac{d}{dx}(3)$, use the Derivative of a Constant Function Rule.

$$\frac{d}{dx}(3) = 0$$

Thus, $\frac{d}{dx}(4x + 3) = 4$.

Now simplify.

$$\frac{(4x + 3)\frac{d}{dx}(5x - 2) - (5x - 2)\frac{d}{dx}(4x + 3)}{(4x + 3)^2} = \frac{(4x + 3)(5) - (5x - 2)(4)}{(4x + 3)^2} \\ = \frac{23}{(4x + 3)^2}$$

$$\text{Thus, } y' = \frac{d}{dx} \left(\frac{5x - 2}{4x + 3} \right) = \frac{23}{(4x + 3)^2}.$$